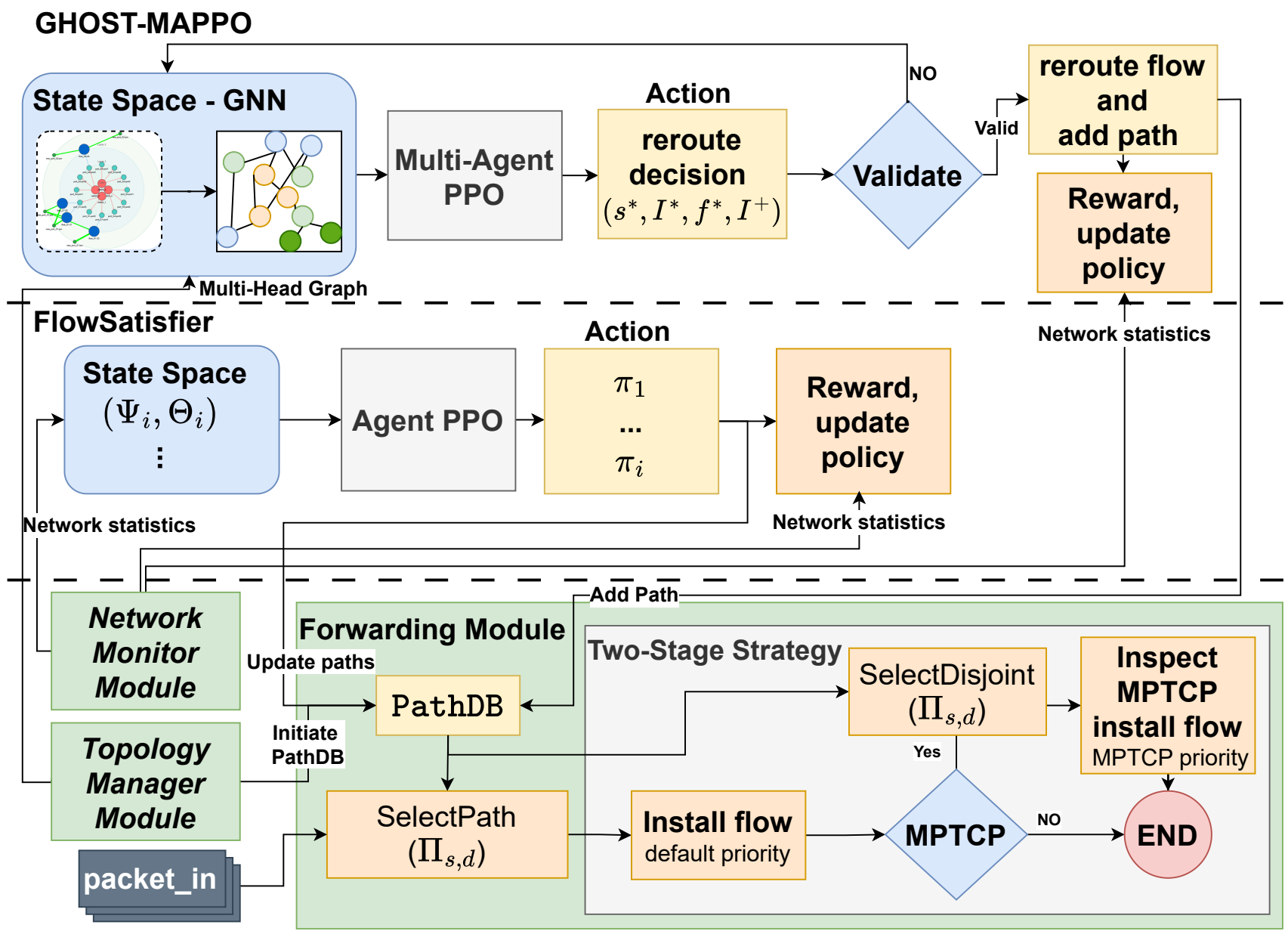
GHOST-MAPPO
State Space - GNN
Multi-Head Graph
Multi-Agent PPO
Action
reroute decision
(s^*, I^*, f^*, I^+)
Validate
NO
Valid
reroute flow and add path
Reward, update policy
Network statistics
FlowSatisfier
State Space
(Ψ_i, Θ_i)
⋮
Agent PPO
Action
π_1
...
π_i
Reward, update policy
Network statistics
Network statistics
Network Monitor Module
Topology Manager Module
packet_in
Forwarding Module
Update paths
Initiate PathDB
PathDB
Add Path
SelectPath
$(\Pi_{s,d})$
Two-Stage Strategy
SelectDisjoint
$(\Pi_{s,d})$
Yes
Inspect MPTCP install flow
MPTCP priority
Install flow
default priority
MPTCP
NO
END